

Capstone Quick Reference: Living Systems as Decentralized Economies

BME 129C — Sage Clokey — Spring 2026, UC Santa Cruz

Main Findings

Layer 1: Network Topology

- E. coli GRN follows power-law (scale-free) degree distribution — no master node
- Betweenness Gini is low — no single gene bottlenecks information flow
- Robust to random failure; centralized (star) networks collapse when hub is removed
- Feed-forward loops massively over-represented (high Z-score vs 1,000 randomizations)
- WBPA self-regulation: hubs don't monopolize, network resists centralization (hub erosion)

Layer 1b: Single-Cell Economy

- PBMC scRNA-seq shows distinct cell-type clusters — same genome, different output
- Shannon entropy varies across cell types — specialists vs generalists = division of labor
- Communication network is distributed — removing any cell type leaves 70-90% of edges
- Same ligand received differently by different cell types — Menger's subjective value

Layer 2: Metabolic Economy

- Distributed agents reach equilibrium without a planner (price discovery then convergence)
- Distributed retains more GDP than centralized after every pathway knockout
- Perturbation suite: distributed wins on substrate shock, ATP crisis, demand spike, novel opportunity
- FBA (iML1515) achieves high but not 100% accuracy — omniscient planner still fails
- Shadow prices shift with context (NADH cheap on glucose, expensive on acetate) — subjective value

Layer 2: Price System (Central Figure)

- Three-tier cellular price system: metabolite ratios, intercellular signals, mTOR integration
- Cancer mutations target price system genes (TCGA, 10,967 samples)
- TP53: 96% ovarian, 1% thyroid
- PIK3CA: 52% uterine, 3% ovarian
- PTEN: 67% uterine, 1% thyroid
- Different tissues break different price components — disease is context-dependent, not variant-intrinsic

Layer 3: Cross-Species Trade

- Organisms specialize — Ricardian comparative advantage (coral: biomineralization, spider: silk, etc.)
 - Trade cost correlates with evolutionary distance — mirrors Tinbergen gravity model
 - Voluntary exchange succeeds, forced exchange fails
 - Trade blocs emerge spontaneously via Louvain clustering — Mengerian spontaneous order
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Barabasi-Albert Model

- Grows networks by adding nodes one at a time with preferential attachment (connect proportional to existing degree)
 - Produces power-law degree distribution: $P(k) \sim k^{-\alpha}$, α between 2-3
 - Null model for decentralized network growth — E. coli GRN matches it
 - Not a star (central planning), not random (no structure), not lattice (no specialization)
 - Capstone extends beyond BA: biology uses WBPA (betweenness-based attachment), which causes hub erosion that standard BA lacks
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Limitations

Data

- Layer 1: E. coli only (one prokaryote) — needs eukaryotic GRNs
- Layer 1b: pbmc3k only (~2,700 cells, blood) — needs multi-tissue data
- Layer 2 simulation: toy model (13 agents, simplified feedback) — demonstrates principle, not full complexity
- Layer 2 FBA: assumes steady state and optimal growth — real cells rarely satisfy both
- Layer 2 price system: conceptual mapping, not direct measurement of cellular prices
- Layer 2 cancer: TCGA correlation, not causation — consistent with thesis, not proof
- Layer 3: only 5 organisms — proof of concept, not comprehensive survey
- Power-law fit debated (Broido & Clauset 2019) — heavy-tailed does not guarantee scale-free

Methodological

- Analogy vs mechanism: data is consistent with economic framework, but consistency is not proof of identical mechanism

- No wet lab validation — entirely computational
- Some alternatives tested (star graph, uniform costs) are straw men — real competition is subtler
- Single-species bias — Layers 1-2 are almost entirely E. coli

What would strengthen it

- Eukaryotic GRN analysis (human, yeast)
 - Multi-tissue scRNA-seq
 - Wet lab validation of at least one prediction
 - 20+ organism trade network with measured expression data
 - Formal comparison against a competing mechanistic model
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Strengths

- **Falsifiable at every layer** — every claim has a figure, a test, and a stated alternative that would disprove it
 - **Multi-scale convergence** — 5 data types, 5 methods, same conclusion (distributed > centralized)
 - **Novel cancer reframing** — TCGA mutations mapped to price system components; tissue-specific variation shows disease is context-dependent
 - **Strongest possible opponent** — FBA comparison uses genome-scale LP with perfect information (iML1515), not a naive planner
 - **Fully reproducible** — `run_all.py` regenerates every figure from source data
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Comparable Publications

1. **Alon (2007)** — "Network Motifs: Theory and Experimental Approaches," *Nature Reviews Genetics* 8, 450-461 - FFL over-representation in E. coli GRN - Capstone Z-scores should match; economic interpretation (FFLs as Hayekian price signals) is novel

2. **Barabasi & Oltvai (2004)** — "Network Biology: Understanding the Cell's Functional Organization," *Nature Reviews Genetics* 5, 101-113 - Scale-free topology, robustness curves, self-organization in biological networks - Capstone degree/robustness data should align; WBPA analysis and economic framework extend beyond their description
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What Makes This Different From the Original Proposal

What the proposal envisioned

- General "spontaneous order in biology" thesis (Hayek, Kauffman, Turing, Prigogine)
- 4 planned layers: GRN topology, metabolic self-organization, HGT, morphogenesis
- Economic analogy was present but loose — cited as philosophical inspiration, not operationalized

What the final research produced that's new

1. **Austrian economics = quantitative framework, not analogy** - Hayek's knowledge problem = 30% FBA failure rate on real knockout data - Mises' calculation problem = 18.1 percentage point GDP gap under perturbation - Kirzner's entrepreneurial discovery = distributed agents respond to novelty before planner can - Menger's subjective value = shadow prices shift across environmental conditions
2. **Price system figure (Panels A-D) is entirely new** - Three tiers: metabolite ratios (cost of capital), intercellular signals (market prices), mTOR (entrepreneur) - Not in the proposal at all — this is the central figure of the paper
3. **Cancer as price system destruction is original** - Proposal said nothing about cancer - Paper maps TCGA mutations (10,967 samples) onto price system components - TP53: 96% ovarian, 1% thyroid — disease is context-dependent, not variant-intrinsic

4. **Immune + genome-wide layers are entirely new (Layers 5-7) - SHM**
hotspots: 19:1 enrichment - V(D)J bias: 10-20x preferred segments,
reproducible across individuals - Public clonotypes: 10^{15} -fold above
random - CpG hotspots: 15-40x baseline - Tissue specificity: 20% of genes
with $\tau > 0.95$, 100-1,000x fold enrichment - Convergent evolution: 35
events, 17 traits, up to 1.5 billion years divergence
5. **FBA reframed as omniscient planner test** - Proposal used FBA as a
metabolic analysis tool - Paper uses it as the strongest central planner and
shows it fails for Austrian reasons
6. **Layer 3 shifted: HGT → Ricardian comparative advantage** - Trade costs,
voluntary exchange success rates, spontaneous trade blocs - More specific
and testable than original "open-source biology" framing
7. **Morphogenesis dropped, replaced with real data** - Turing pattern
simulation replaced by measured immune/genomic data - Stronger evidence:
real data > computational simulation

Bottom line

- Proposal: "biology looks like spontaneous order"
 - Paper: "biology operates as a decentralized economy with a measurable price system — here are the numbers, and here's what happens when it breaks"
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Life as the Original Internet / Adaptive Computer

- Life is the original internet — decentralized, self-assembling, adaptive, 4 billion years before the first packet was routed
- The parallels are architectural, not metaphorical — the capstone data demonstrates each one

Routing around damage

- Internet: packets route around failed servers
- Biology: PPI networks survive 37% node removal; star graph collapses at 1.9% (19:1 ratio)

- Same structural advantage, biology discovered it first

No master server

- Internet: no master router, every router knows only its neighbors
- Biology: power-law topology, distributed betweenness, hub erosion via WBPA
- Internet uses BGP routing tables; biology uses betweenness-preferential attachment

Local knowledge, global coordination

- Internet: routers make local decisions, packets reach any destination globally
- Biology: metabolic agents read local pools, economy reaches equilibrium without a planner
- Both solve Hayek's knowledge problem through distributed architecture
- Oscillations during convergence = price discovery = routing convergence

Self-assembling

- Internet was engineered to be decentralized — humans designed the protocols
- Life grew that way — no engineer, no blueprint, self-assembly from local rules
- 37 trillion cells differentiate from one genome without a master cell assigning roles
- Life is the internet that built itself

Adaptive computation

- Every cell is a computational node — mTOR integrates multiple price signals into grow/conserved decisions
- 37 trillion processors running in parallel, each computing locally
- Internet: distributed computation, no single CPU runs the network
- Biology: distributed molecular computation, no single cell runs the organism

The deeper point

- Humans didn't invent decentralized networks — we recognized the one already running
 - Internet, blockchain, mesh networking = recapitulations of an architecture life has run for 4 billion years
 - Same structural properties (power-law, robustness, local coordination, self-organization) because same fundamental problem: coordinating autonomous agents without central control
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Significance of the Study

- **Paradigm challenge:** dominant framework (iGEM, BioBricks, CRISPR, precision medicine) treats life as a machine — parts to swap, circuits to wire
 - This study shows that assumption is structurally wrong — not a data gap, a framework error
 - The function is in the connections, not the parts — a gene is an economic agent in a network, not a machine component
 - TP53: 96% in ovarian, 1% in thyroid — the part didn't change, the network did
 - Machine paradigm produces fragile designs: circuits fail in new contexts, codon optimization destroys information, drug targets fail in trials
 - **First unified, quantitative, multi-scale demonstration** that biology operates as a decentralized economy with falsifiable predictions at every layer
 - The shift: stop designing from above, start cultivating from within; stop treating variation as defect, start treating it as raw material for adaptation
 - **Living things are not machines. They are economies.**
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Unanswered Questions and Future Research

1. **Eukaryotic GRNs** — Does power-law, hub erosion, WBPA hold in human/yeast networks? Data exists (ENCODE, JASPAR) but needs extraction at same detail level
 2. **Direct price system measurement** — Can ATP/ADP, cytokine levels, and mTOR activation be tracked in real-time in the same cell under perturbation? Single-cell metabolomics approaching this resolution
 3. **Wet lab prediction test** — Does knocking out a price receptor (EGFR) or integrator (PTEN) in an organoid produce centralized-economy behavior? (reduced specialization, lost division of labor, fragile perturbation response)
 4. **Predictive trade cost model** — Can codon distance + regulatory barrier scores predict heterologous expression success before the experiment? Systematic screen of 50+ transfers would test this
 5. **Immune distributed knowledge in disease** — Does autoimmunity = misreading self-signals (broken subjective value)? Does immunodeficiency = lost entrepreneurial discovery (reduced V(D)J diversity)?
 6. **Disease threshold** — Is there a measurable tipping point (Gini, centrality, shadow price distortion) where distributed shifts to centralized? Could enable early detection before clinical disease
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Consequences If the Problem Is Avoided

Synthetic biology fails at scale

- Single-pathway mods work; multi-pathway integration fails
- Machine paradigm says: better parts. Distributed framework says: understand the connections
- Designing machines when you should be cultivating economies

Drug development keeps failing

- Single-target model has >90% clinical trial failure rate

- Target is not a part, it's a node in a distributed economy — silencing one node doesn't fix the system
- Network reroutes, compensates, or collapses elsewhere

Genomics misinterprets variants

- Machine logic: variants = defects — same logic that powered eugenics
- Data shows: same variant is catastrophic in one context, irrelevant in another
- Disease is not the variant — disease is the broken context (price system that stopped reading the signal)
- Variants are not diseases. An elephant is not diseased because it can't climb a tree.

Cancer research chases parts instead of systems

- Dominant approach: identify mutated gene, target it individually
- Cancer is destruction of a distributed price system — tissue-specific, multi-component
- Targeting one gene when multiple price tiers are disrupted = fixing one traffic light when the entire road network is rewired

The deeper consequence

- Machine metaphor trains engineers to think like central planners
 - Produces biologists who design from above, see variation as error, treat self-organization as obstacle
 - The fatal conceit: believing human reason can redesign what distributed processes built over 4 billion years
 - **Biology is not a machine. The connections are the function, not the parts. The sooner the field recognizes this, the sooner it stops building systems that break.**
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Connection Is What Makes Life — Isolation Does Not Keep People Safe

The immune system depends on connection

- V(D)J recombination = diversity through combinatorial contact
- Somatic hypermutation = improvement through iterative exposure and feedback
- Public clonotypes = independent immune systems converge on same solutions (10^{15} -fold above random)
- The immune system doesn't get stronger through isolation — it learns through encounter
- Sever the contact and you don't protect the system, you prevent it from learning

Network data: resilience = connection

- Layer 1b: remove a cell type, 70-90% of edges survive — robustness is in the paths, not the nodes
- Remove the connections themselves and the network collapses
- 19:1 robustness ratio is about how many paths remain, not which nodes you keep
- Healthy nodes with severed connections = fragile network

Metabolic economy requires communication

- Layer 2: distributed agents recover because they read shared metabolite pools (prices)
- Isolation = cutting agents off from the pool — they can't read prices, can't adapt, can't discover equilibrium
- Centralized planner fails because it assigns quotas instead of reading local conditions

Central planning applied to public health

- COVID lockdowns = one authority deciding who connects, when, how

- Distributed framework predicts the result: severing connections makes networks fragile, not safe
- People cut off from social, economic, immunological connections
- Small businesses (distributed agents) shut down; centralized institutions kept operating
- Local knowledge overridden by top-down mandate treating every person as interchangeable

The lie and the truth

- The lie: safety comes from separation
- The data: connection is what makes life, at every scale
- Immune system learns through exposure
- Metabolic economy coordinates through shared signals
- GRN survives through distributed paths
- Cell communication functions through voluntary exchange
- Isolation severs the connections that make resilience possible
- The network that routes around damage survives; the network that preemptively severs its own links has nothing left

How to Do Things WITH Life Instead of Using Life as a Machine Part

The wrong question vs the right question

- Wrong: "How do we program life?"
- Right: "How do we join the computation?"
- Life is already computing — 37 trillion nodes, 4 billion years, no central CPU
- Lead life to lead itself — be a node that makes an effect, not a planner that commands

Be a node, not a planner

- Layer 1: no master node. Layer 2: distributed beats omniscient planner
- Engineer who controls from above = star graph = 19x more fragile
- Engineer who enters as a node = participates in architecture that already works
- Don't need to understand the whole system — act on local conditions, network routes the contribution

Cultivate conditions, don't command outcomes

- Layer 2 price discovery: agents find equilibrium through local feedback, nobody told them what to produce
- Engineering move: set conditions, provide substrate/signals/environment, let distributed computation run
- The invisible hand doesn't need instructions — it needs conditions

Reduce trade barriers, don't force trade

- Layer 3: voluntary exchange succeeds, forced exchange destroys information
- Codon harmonization (reduce friction) works; forced optimization (command from above) breaks folding
- Enable trade between compatible systems, don't force genes into incompatible hosts

Read the price system, don't override it

- Cells already have a 3-tier signaling economy — mTOR, cytokines, morphogens
- Work with existing signals: add what the system can interpret, remove barriers it struggles against
- Constitutive promoter + synthetic circuit = central planner ignoring prices and assigning quotas

Steve Jobs and the bridge from silicon to carbon

- Jobs didn't invent computation — he made it accessible (mainframe → personal → pocket)
- Moved computation from center to edge, institution to person, planner to node
- "Everything around you was made by people no smarter than you — you can change it, build your own things that other people can use"
- That's the node philosophy: enter the network, contribute, let it route your work where it's needed
- Jobs created silicon life → accessible computing → enabled genomics, FBA, scRNA-seq, this capstone
- Silicon computation was the prerequisite for understanding biological computation
- Next step: work with biological computation directly

The Living Age

- Bridge from silicon to carbon, programming to cultivating, central planning to distributed participation
- Not designing organisms from above — entering the biological economy as a participant
- Read price signals, reduce trade barriers, cultivate conditions, lead life to lead itself
- The computation of life is not something to invent — it's something to join
- Stop being central planners, start being nodes — gain access to 4 billion years of adaptive intelligence